

Madeleine Wallace

PHD CANDIDATE · SCHOOL OF NATURAL RESOURCES AND THE ENVIRONMENT

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Education

University of Arizona

PHD CANDIDATE, NATURAL RESOURCES (ECOLOGY AND MANAGEMENT OF RANGELANDS)

- Minor in Science Communication
- Advisor: Dr. Rachel M. Mitchell

Tucson, AZ

2023 - present

University of Portland

BS ENVIRONMENTAL SCIENCE (BIOLOGY)

- Research advisor: Dr. Heather Carpenter
- Thesis: *Anthropogenic effects on the killer whales (Orcinus orca) of Puget Sound*

Portland, OR

2019

Professional Experience

- 2023-Pres **Graduate Research and Teaching Assistant**, UA School of Natural Resources and the Environment
- 2025 - Pres **Community Action for Biodiversity Fellow**, UA Lovejoy Center
- 2024 - Pres **Outdoor Educator**, UA Mt. Lemmon Sky School
- 2020-2023 **Project Manager**, Native Plantscapes Northwest, Hillsboro, OR
- 2017-2019 **Life Sciences Educator**, Oregon Museum of Science and Industry, Portland, OR
- 2017-2019 **Undergraduate Research and Teaching Assistant**, UP Biology Department

Publications

PEER REVIEWED

- Winick, I., **Wallace, M.**, Bromley, F., Martinez, C., Crowder, C., Goolsby, D., Taber, E. M., Mitchell, R. (2026). High-Severity Fire Selects for “Fast” Traits and Distinct Plant Communities 5 Years After a Mixed-Severity Fire. *Journal of Vegetation Science*, 37(3), e70157. <https://doi.org/10.1111/jvs.70157>.
- Diepenheim, G., Gift, S.C., Harb, C., **Wallace, M.**, Layshock, J. (2020). Survey of phthalate mitigation and distribution in water, sediment, and *Typha* spp. in a fully operational constructed wetland: a pilot study. *Bulletin of Environmental Contamination and Toxicology*, 105: 205-210. doi:10.1007/s00128-020-02915-9.

IN REVIEW

- Wallace, M.**, Mitchell, R.M. Transgenerational plasticity in response to water limitation leads to reductions in fitness in *Plantago patagonica*, an arid-adapted annual plant. *Oecologia*.
- Javier, V., **Wallace, M.**, Goolsby, D.C., Mitchell, R.M. *Oncosiphon pilulifer*, a rapidly expanding invader, exhibits local adaptation to dry conditions. *AoB Plants*.

NON-PEER REVIEWED

- Wallace, M.** (2025, May 15). Home on the range: student researcher sees beauty in desert field work. *Green Valley News*.

Awards, Fellowships, & Grants

2025-2027	Community Action for Biodiversity Fellowship , UA Lovejoy Center	\$20,000
2025-2026	Carson Scholar , UA Arizona Institute for Resilience	\$5,000
2025	Best Graduate Poster, 1st Place , UA Research Insights in Semi-Arid Ecosystems Symposium	\$500
2024-2027	Provost Investment Fund, with R. M. Mitchell , UA Office of the Provost	
2024	Leadership Award , UA School of Natural Resources and the Environment	
2024	Travel Grant , ESA Population Ecology Section	\$300
2020	Best Poster, 2nd Place , Oregon Academy of Science & Sigma Xi Research Symposium	\$250
2019	Butine Grant , University of Portland	\$1,460

Presentations

CONTRIBUTED PRESENTATIONS

Wallace, M., Mitchell, R.M. 2025. *Generating drought resilient native plant materials: A case study using Plantago patagonica, a US southwestern restoration species*. Oral presentation: Society for Ecological Restoration Annual Conference, Denver, CO.

Wallace, M., Mitchell, R.M. 2025. *Seventy Years of Change: Climate and Vegetation Dynamics on the Santa Rita Experimental Range*. Poster: Research Insights in Semi-Arid Ecosystems Symposium, Tucson, AZ.

Wallace, M., Mitchell, R.M. 2024. *Transgenerational plasticity in response to drought affects functional traits of Plantago patagonica, a priority restoration species*. Poster: School of Natural Resources and the Environment 50th Anniversary Symposium, Tucson, AZ.

Wallace, M., Mitchell, R.M. 2024. *Transgenerational plasticity in response to drought affects functional traits of Plantago patagonica, a priority restoration species*. Poster: Ecological Society of America Annual Meeting, Long Beach, CA.

Wallace, M., Mitchell, R.M. 2024. *Transgenerational plasticity in response to drought affects functional traits of Plantago patagonica, a priority restoration species*. Poster: Agriculture, Life, Veterinary Sciences, and Cooperative Extension Annual Research Poster Showcase, Tucson, AZ.

Wallace, M., Mitchell, R.M. 2023. *Seed origin and drought affect offspring life history traits in Plantago patagonica*. Oral presentation: Society for Ecological Restoration Southwest Chapter Annual Meeting, Santa Fe, NM.

Wallace, M., Mealy, M., She, S., Gima, C., Dizney, L. 2020. *Cardenolide content in native Oregon milkweed under various environmental conditions*. Poster: Oregon Academy of Science & Sigma Xi Student Research Symposium, Portland, OR.

Wallace, M., Mealy, M., She, S., Gima, C., Dizney, L. 2019. *Cardenolide content in native Oregon milkweed under various environmental conditions*. Poster: University of Portland Annual Fall Research Poster Session, Portland, OR.

Wallace, M. 2018. *Anthropogenic effects on the killer whales (Orcinus orca) of Puget Sound*. Oral presentation: University of Portland Annual Founder's Day Symposium, Portland, OR.

PUBLIC & OUTREACH PRESENTATIONS

Wallace, M. 2026. *Sonoran Desert native plants*. Public science presentation: Tumamoc Hill Youth Program, Desert Laboratory on Tumamoc Hill, University of Arizona, Tucson, AZ.

Teaching Experience

INSTRUCTOR OF RECORD

SU2026 **RNR 160: Wildlife, Conservation, and Culture**, SNRE

SU2025 **RNR 160: Wildlife, Conservation, and Culture**, SNRE

SU2024 **RNR 150: Sustainable Earth: Natural Resources and the Environment**, SNRE

GRADUATE TEACHING ASSISTANT

SP2024 **RNR 321: Ecological Surveys and Sampling**, SNRE

SP2023 **RNR 321: Ecological Surveys and Sampling**, SNRE

FA2023 **RNR 150: Sustainable Earth: Natural Resources and the Environment**, SNRE

UNDERGRADUATE TEACHING ASSISTANT

- FA2019 **BIO 472: Ecology Laboratory**, Department of Biology
- SP2018 **CHM 375/376: Organic Chemistry Laboratory**, Department of Chemistry
- FA2018 **CHM 207/208: General Chemistry Laboratory**, Department of Chemistry
- SP2017 **MTH 361: Applied Statistics**, Department of Mathematics

Outreach & Professional Development

SERVICE AND OUTREACH

- 2025-Pres **Tierra Seca Rangeland Club**, Graduate student advisor
- 2025-Pres **UA Mt. Lemmon Sky School**, Outdoor educator
- 2024-Pres **SNRE Inclusive Excellence Committee**, Graduate student representative
- 2024-2025 **SARSEF Regional Science Fair**, Biology judge
- 2023-Pres **Girls on outdoor Adventure for Leadership and Science (GALS)**, Trip leader
- 2023-2024 **UA Earth Week Organizing Committee**, Co-chair
- 2023-2024 **SNRE Graduate Student Organization**, Co-president

PROFESSIONAL DEVELOPMENT

Transdisciplinary Research Academy (2026), ten sessions that introduced key skills for successful transdisciplinary research through the Office of Societal Impact and Arizona Institute for Resilience at UA.

Carson Scholars (2025-2026), a year long science communication training fellowship with the Arizona Institute for Resilience at UA.

Reproducibility and Data Science Skills Workshop (2023), ten sessions covering Git and GitHub, project management, coding best practices, and data cleaning and visualization in R with the Data Science Team at UA Libraries .

PROFESSIONAL MEMBERSHIPS

Society for Range Management, Society for Ecological Restoration, Ecological Society of America

Skills

FIELD Wilderness First Aid (NOLS); plant surveys and rangeland monitoring; plant identification

SOFTWARE & CODING R / RStudio; Git / GitHub; Markdown; Quarto; ArcGIS Pro

PROJECT MANAGEMENT Restoration project design and implementation; landowner recruitment and needs assessment; budgeting and effectiveness monitoring; crew and client management; stakeholder reporting; long-term land management planning

COLLABORATION Cross-sector partnership with nonprofits, state and federal agencies, and industry; communicating science to technical, management, and public audiences; interdisciplinary, team-based research and development of restoration projects